Supplemantry Material for jrfm-4129983

2.2 Search Strategy and Protocol

In order to guarantee academic rigor, replicability, and methodological transparency, this research employs the PRISMA framework to guide the SLR process during Stage 1. The objective is to identify prior review-based studies focusing on modeling and prediction within financial markets, thereby establishing a strong theoretical foundation for the subsequent BA and CA.

2.2.1 Database Selection

The WoS Core Collection was chosen as the sole data source over Scopus and Google Scholar for several reasons: (1) Historical coverage: WoS includes records back to 1900, versus 1960 for Scopus, offering a broader temporal scope (Mongeon & Paul-Hus, 2016). (2) Output volume: A search for "FMM" in WoS retrieved more relevant articles (8,644 records since 1931) than Scopus (4,837 records since 1969). (3) Quality and indexing: WoS restrict coverage to peer-reviewed journals from major publishers (e.g., Elsevier, Wiley, Springer Nature, Taylor & Francis, Oxford University Press), ensuring methodological rigor and rich citation data (Li, Rollins & Yan, 2018; Liu, 2017).

Standardization and Software Integration: The platform provides well-structured metadata compatible with Bibliometrix/Biblioshiny and EndNote, enabling accurate data exports and high replicability (Aria & Cuccurullo, 2017; Prado et al., 2016). Disciplinary Fit: Web of Science (WoS) covers a wide range of fields—finance, business, computer science, and economics—making it suitable for this interdisciplinary study. Prior Precedent: Many bibliometric and SLR studies confirm WoS as a reliable standalone source for finance and modeling research (e.g., Prado et al., 2016). Although Scopus is widely used for bibliometric analyses (Donthu et al., 2021; Paul et al., 2021), WoS is preferred here for its broader topic-specific coverage and greater bibliometric precision.

2.2.2 Search Strategy: Field-Specific Boolean Query Logic

An advanced Boolean query was used across selected fields to optimize recall and precision. Review-related terms were searched in the Title field (TI=) to limit results to works explicitly categorized as reviews. Terms related to models and financial markets were searched in the Topic field (TS=), retrieving matches from titles, abstracts, author information, keywords, and Keywords Plus.

Review –related keywords: ("Review" OR "Bibliometri*" OR "scientometric*" OR "Systematic Review" OR "Future Trend*" OR "synthes?s" OR "literature*" OR “Content Analysis”) AND
Models –related keywords: (“Model*” OR “Predict*” OR “forecast*”) AND
Financial Markets – related keywords: (“Financial Market*” OR “Securit* market*” OR “Capital market*” OR “Stock* Market*” OR “Stock Exchange” OR “Bond* market*” OR “Commodit* Market*” OR “Energy Market*” OR “Derivatives market*” OR “Foreign exchange*” OR “$currenc*")

2.2.3 Filtering Criteria

To ensure methodological precision and relevance, the search was limited to Review Articles or Articles with "apply bibliometric" in the WoS Core Collection (excluding the Preprint Citation Index). The research areas were Business Economics, Engineering, Computer Science, Energy Fuels, Mathematics, Environmental Sciences & Ecology, Physics, and Science Technology Other Topics. Only English-language publications from 1990 to 2024 were included.

2.2.4 PRISMA Protocol Implementation

A four-phase review procedure based on Page et al. (2021) was applied: (1) Identification: the initial query yielded 928 documents. (2) Screening: titles and abstracts were examined to remove irrelevant items, including preprints, purely empirical studies, and off-topic research. (3) Eligibility: 444 full-text articles were assessed for thematic and methodological fit; studies without substantive financial modeling or focused solely on non-financial sectors were excluded. (4) Inclusion: 67 high-quality, review-oriented papers on financial market modeling or forecasting were retained (e.g., systematic

reviews, meta-analyses, bibliometric reviews) that met criteria on publication type, language, and subject. This targeted strategy grounded Stage 1 in high-quality secondary literature in line with PRISMA guidelines, validated and contextualized the research design, and provided a conceptual framework for Stages 2 and 3, with key insights detailed in the Market-Specific Reviews and Key Findings sections and summarized in the Supplementary materials.

Table S1
Previous Literature Reviews including our current paper.

NO.	Market	Manuscript	No. of Publications	Database	Period	Aim of Study	Type of Method	Main Conclucions
1	**Financial Markets**	Poon & Granger (2003)	93	N/A	1976 - 2002	Provides a comprehensive review of 93 studies and ranks volatility forecasting models in financial markets.	Systematic Review Analysis	HISVOL and GARCH offer similar accuracy, with ISD performing best. Implied volatility uses richer data but lacks accessibility; historical methods match ARCH and SV models.
2	**Financial Markets**	Bahrammirzaee (2010)	Not disclosed / Specified But approx. 201 articles (counted)	Not disclosed / Specified	1989 - 2009	Reviews and compares ANN, expert systems, and hybrid AI techniques in financial markets over two decades, focusing on their performance versus traditional and other intelligent methods.	comparative research review	AI methods generally outperform traditional models, especially for nonlinear patterns. NNs show strong predictive power despite issues like overfitting and slow training. ES lacks adaptability. Overall, AI offers superior accuracy, particularly in unstructured financial contexts.
3	**Financial Markets**	Maiti (2020)	Not disclosed / Specified	Not disclosed / Specified	Not disclosed / Specified	Critically reviews the evolution of risk factors and factor models through a structured literature review, identifying key trends, contributions, limitations, and future research directions.	Critical Review Analysis	The Fama-French five-factor model marks progress but doesn't settle asset-pricing debates. Factor models work in specific markets but lack global consistency, as evolving risk factors prevent stable, universal application.
4	Financial Markets **"FOCUS MORE ON STOCK MARKETS" - Price, return, movements, volatility, … etc**	Henrique, et al., (2019)	57	Scoups	**1991 - 2017**	Applies bibliographic survey methods to identify key studies on machine learning for financial (stock) market prediction, offering a focused review of influential literature in the field	Bibliometric Survey technique	Customized ML models enhance prediction accuracy. Neural networks and SVMs dominate, while other models see limited use. Evaluation varies by forecast type—directional accuracy or error metrics.

5	**Financial Markets "Mutual Funds"**	Elton & Gruber (2020)	Not disclosed / Specified	Not disclosed / Specified	Not disclosed / Specified	Reviews the extensive literature on mutual fund performance evaluation developed to guide investment decision-making.	Simple Review "Viewpoint"	Active funds often underperform passive benchmarks. Factor models and APT help evaluate mutual fund performance, though benchmarks may lack long-term economic stability.
6	**Financial Markets**	Chen & Yang (2021)	189	Web of Science (WOS)	2001 - 2017	Uses bibliometric mapping to identify the knowledge base and research trends in cross-market volatility spillovers.	Bibliometric Review Analysis	Volatility spillover research centers on multivariate GARCH and VAR models, focusing on spillover dynamics, oil market volatility, and emerging market risks. Studies highlight long memory, asymmetry, and structural shifts, with strong collaboration among U.S., European, and Chinese scholars.
7	**Financial Markets**	Nassirtoussi, et al., (2014)	Not disclosed / Specified	Not disclosed / Specified	Not disclosed / Specified	Reviews market prediction models using online text mining, outlining common components and comparing systems to highlight key differentiators	Systematic Review Analysis and Comprehensive comparative analysis	Market prediction via text mining shows 50–70% accuracy, with 55% often deemed significant. Feature selection is vital for high-dimensional data, though varied evaluation methods hinder comparison. The field is growing with tech advancements.
8	**Financial Markets**	Cavalcante, et al., (2016)	56	Science Direct, Google Scholar and IEEExplore digital libraries	2009 - 2015	Reviews recent computational intelligence methods applied to financial market problems.	Comprehensive Survey	Financial data complexity favors non-parametric and hybrid models. While ARIMA and GARCH remain useful, many intelligent forecasting studies lack practical risk or profit evaluation. Technical analysis is widely used; fundamental analysis and text mining face challenges in real-world application.
9	**Financial Markets**	Kumar & Ravi (2016)	89	ACM digital library, Taylor and Francis, Science Direct, Wiley, Google Scholar, Springer and IEEE-Xplore	2000 - 2016	Presents a comprehensive review of text mining applications in finance, including stock and FOREX prediction, and cybersecurity tasks like fraud and phishing detection.	Comprehensive Survey	Using full news articles enhances stock prediction models. As text data grows, advanced extraction methods and feature reduction are key. SVM dominates classification tasks, though fraud detection remains underexplored. Evaluation methods vary widely.
10	**Financial Markets**	Baillie (1996)	138	Not disclosed / Specified	Not disclosed / Specified	Reviews econometric research on long memory and fractionally integrated processes, focusing on their applications in economics and finance, with selective coverage of relevant statistical literature.	Survey and Review	Forecast accuracy varies with autoregression order and horizon. Long-memory models like FIGARCH improve flexibility and capture volatility and return dynamics better than unit root models.
11	**Financial Markets "Stcok Markets & FOREX" Prediction**	Xing, et al., (2018)	Not disclosed / Specified	Not disclosed / Specified	1998 - 2016	Reviews and organizes NLP-based financial forecasting (NLFF) techniques, highlighting research progress, key applications, and interdisciplinary developments.	Survey	Realized volatility impacts wealth and trading strategies. Sentiment reversals guide trades but have weak predictive power. Weekly sentiment patterns and disagreements influence market returns and trading volume, especially in developed markets.

12	**Financial Markets**	Jiang, et al., (2019)	Not disclosed / Specified	Not disclosed / Specified	Not disclosed / Specified	Reviews multifractal models applied to financial time series, highlighting their properties, sources, and role in market inefficiency analysis, risk management, and broader applications, while outlining open research directions.	Survey	Multifractal analysis faces challenges in method and scaling range selection, especially with short time series. Current criteria lack standardization, and distinguishing true sources of multifractality remains essential for accurate complexity assessment.
13	**Financial Markets & Banking Sector**	Sanchez-Roger, et al., (2019)	795	Web of Science, Scopus and, Google Scholar	**1987 - 2017**	**Critically examines fuzzy logic as a valuable tool in financial research, with emphasis on its application to managing banking crises.**	Systematic and Bibliometric Review Analysis	The fuzzy logic–neural network hybrid is widely used and effective, especially in banking crises, trading, and fraud detection. Its adaptability to complex financial data makes neuro-fuzzy systems particularly efficient for transparency and risk analysis.
14	**Financial Markets "Stcok Markets & Cryptocurrency"**	Nosratabadi, et al., (2020)	57	Thomson Reuters Web of Science citation database and Elsevier Scopus	2013 - 2019	Presents a comprehensive review of recent advances in data science applied to emerging economic domains, identifying key application areas and popular modeling techniques.	Systematic Comprehensive Review	Hybrid deep learning models outperform traditional methods in stock prediction, with LSTM, CNN, and DNN widely used. Despite advancements, few studies utilize financial news or social media. NLP-based forecasting is emerging, with hybrids showing superior accuracy through lower RMSE.
15	**Financial Markets**	Narayan & Smyth (2015)	Not disclosed / Specified	Not disclosed / Specified	Not disclosed / Specified	Reviews recent econometric advances in price discovery and predictability, addressing traditional model limitations, newer methods, current trends, and future research directions.	Systematic Review	Advances in econometrics have enhanced price discovery research, highlighting home and futures markets as key sources. Crises and speculation introduce nonlinear effects. Fractional cointegration models outperform traditional VECMs in multi-market analysis, improving accuracy and addressing over-parameterization.
16	**Financial Markets**	Chen, et al., (2012)	Not disclosed / Specified	Not disclosed / Specified	Not disclosed / Specified	Reviews the relationship between agent-based computational economics (ACE) and econometrics, noting that most research flows from econometrics to ACE, strengthening its empirical foundations.	Survey	Simple agent-based models, especially fundamentalist–chartist types, replicate market behavior effectively. Estimation reveals key traits, but validation is limited by parameter complexity. Majority switching dynamics support simpler over multi-type agent models.
17	**Financial Markets "Stock markets, foreign exchange markets, and Futures Markets"**	Park & Irwin (2007)	137	Not disclosed / Specified	1960 - 2004	Comprehensively reviews empirical studies on technical analysis, evaluating the consistency and reliability of trading profits across different markets and time periods.	Survey	Technical trading often outperforms buy-and-hold, especially in emerging markets, though profitability depends on costs. Returns are skewed, making t-tests unreliable. Volume enhances signal quality, and pattern-based rules show consistent success across asset classes, unlike optimized rules in developed markets.
18	**Financial Markets "Stock markets, foreign exchange markets"**	Huang, et al., (2007)	Not disclosed / Specified	Not disclosed / Specified	Not disclosed / Specified	Highlights the limited focus on input selection for neural networks, emphasizing its critical impact on forecasting performance and calling for deeper discussion of input variable design.	Survey	ANNs show promise in financial forecasting but face instability and mixed out-of-sample performance. VAR models effectively capture variable interdependencies, with historical data and U.S. market influence aiding prediction. ANNs are more reliable for long-term forecasts than short-term ones

19	**Financial Markets "Case study on forecasting of Cryptocurrencies"**	Blackledge & Lamphiere (2021)	Not disclosed / Specified	Not disclosed / Specified	Not disclosed / Specified	Provides an overview of the historical development of financial time series modeling and reviews related literature on model evolution	Review	Financial markets combine efficient and inefficient traits. Traditional models assume mild randomness, but fractal-based approaches better capture long-term dependencies and extreme behaviors under the Fractal Market Hypothesis.
20	**Financial Markets**	Behera, et al., (2023)	Not disclosed / Specified	Google Scholar	2007 - 2023	Presents a comprehensive review of HONN-based financial forecasting, focusing on the role of evolutionary optimization techniques. Highlights their application across various assets, including stocks, oil, gold, and cryptocurrencies, and emphasizes the reliability and potential of HONN–EOA hybrids.	Survey	HONNs outperform traditional ANNs in forecasting accuracy, speed, and adaptability. When combined with EOAs, they achieve superior performance by optimizing parameters and capturing complex financial patterns.
21	**Financial Markets*****	Nazareth & Reddy (2023)	126	Science Direct	2015 - 2023	This review contributes by analyzing data features used in model training, assessing validation methods, and evaluating model performance across various financial problems.	Systematic Reviews and meta-Analyses.	ML has transformed stock forecasting, outperforming traditional models with tools like LSTM, SVM, and RF using diverse data inputs. While effective in portfolio management, ML remains underused in crypto and crisis prediction, with deep learning applications notably lacking.
22	**Stock Market*****	Bustos & Pomares-Quimbaya (2020)	53	Web of Science, and Scopus	2014-2018	Provides an updated systematic review of stock market forecasting techniques, focusing on their classification, features, and comparative performance in predicting market movements.	systematic review and survey review analysis	Technical indicators are highly predictive inputs. Ensemble models outperform SVM and ANN, while deep learning shows mixed results—likely due to limited dataset adequacy.
23	**Stock Market*****	Thakkar & Chaudhari (2021)	31	Google Scholar	2017 - 2020	Reviews the role of DNNs in stock price and trend prediction, highlighting their applicability to temporal data and exploring hybrid and metaheuristic DNN approaches.	A focused survey on the recent advances of DNNs for stock market prediction.	Highlights advances in applying DL-based neural networks to stock prediction, emphasizing their strength in modeling non-linear, temporal patterns and the need to balance financial theory with computational intelligence.
24	**Stock Market**	Ferreira, et al., (2021)	2326	SCOPUS	1995 - 2019	Analyzes AI's evolution in portfolio optimization, asset price prediction, and sentiment analysis, tracing each area from early applications to current advancements.	Systematic Review Analysis	Portfolio optimization increasingly relies on multiobjective and heuristic models, with deep learning gaining traction. However, comparisons with simpler methods are limited, and feature selection varies widely across studies.
25	**Stock Market**	Kumar, et al., (2021)	84	Not disclosed / Specified	No Range	Reviews recent computational intelligence methods for stock market forecasting, offering a systematic framework to guide researchers, analysts, and investors in developing intelligent prediction models.	Survey	Technical indicators are key to stock forecasting, but optimal selection is challenging. Preprocessing and hybrid models improve accuracy, though evaluation metrics remain inconsistent across studies.

26	**Stock Market**	Sharma, et al., (2020)	307	Web of Science	1960 - 2017	Offers a conceptual synthesis of AI literature in financial prediction, encouraging the use of ML models like MLP, SVM, and LSTM to address nonlinear data in return forecasting, risk modeling, and portfolio construction.	Systematic Review Analysis	AI and neural networks improve nonlinear forecasting, but gaps remain in theoretical generalization and model development, particularly in developing countries compared to advanced economies.
27	**Stock Market**	Nti, et al., (2020)	122	NOT STATED	2007 - 2018	Provides a systematic review of stock market prediction studies from technical and fundamental perspectives, highlighting key influencing factors, comparing predictive methods, and outlining future research directions to address existing gaps.	systematic and critical review analysis	The review shows a dominance of technical analysis (66%) and single-source data (89%). ANN and SVM are most used, with SMA, EMA, and RSI as key indicators. Models often outperform benchmarks, though integrating behavioral and fundamental factors remains a key gap.
28	**Stock Market*** (news)**	Usmani & Shamsi (2021)	99	Not disclosed / Specified	1999 - 2021	Presents a comprehensive review of stock trend prediction using financial and textual data, covering preprocessing, feature extraction, and prediction algorithms. It also identifies challenges, proposes solutions, and outlines future research directions.	Survey	Provides a structured review of stock trend prediction using news and price data, outlining a three-phase framework, key forecasting approaches, and current challenges with proposed solutions in news-sensitive modeling.
29	**Stock Market**	Ye & Li (2017)	Not disclosed / Specified	CNKI and Web of Science	2010 - 2015	Summarizes internet big data sources and analytical methods used in capital market research, reviews key findings, and offers future research directions for applying big data in financial studies.	Structured Literature Review	Categorizes internet big data into forums, microblogs, and search data, using sentiment analysis for market prediction. Despite promising accuracy (~70%), results are inconsistent due to sample and replication issues. Big data shows potential for forecasting markets and company performance.
30	**Stock Market *** LSTM**	Li & Bastos (2020)	34	Scopus, Web of Science (WoS) and IEEE Xplore databases	No Range	Systematically reviews literature on deep learning techniques for stock price forecasting, emphasizing model accuracy, profitability metrics, and adopted trading strategies	Systematic Review Analysis	LSTM dominates stock forecasting (73.5%) for its memory strength, but few studies assess profitability or risk. Technical indicator usage lacks standardization. Hybrid models and basic trading strategies are common, but more advanced methods are needed for volatile, long-term forecasting
31	**Stock Market***Macro**	Verma & Bansal (2021)	201	Emerald Insight EBSCO, Science Direct, and JSTOR	1972 - 2021	Examines the impact of macroeconomic variables on stock market performance across countries and indices, explores stability of these effects, and identifies gaps—particularly in underexplored sectors within the Indian context	Systematic Review Analysis	Macroeconomic variables like GDP, FDI, and FII positively influence stock markets, while gold and interest rates typically have negative effects. Oil price impact varies by country type, and some variables affect markets broadly, while others show sector-specific effects.
32	**Stock Market**	Nardo, et al., (2026)	Not disclosed /	Not disclosed / Specified	Not disclosed /	nvestigates the influence of online news on financial markets, reviewing literature with mixed evidence on its predictive power for market movements.	Survey	Web-based indicators offer modest predictive gains, yet remain valuable in finance. Advanced models don't always improve outcomes. Sentiment effects—like overpricing and herding—are more pronounced after major events, with online activity

			Specified		Specified			metrics aiding analysis of bubbles and market behavior.
33	**Stock Market**	Koundouri, et al., (2016)	Not disclosed / Specified	Not disclosed / Specified	1960 - 2016	Surveys historical statistical models of stock returns and evaluates them as explanatory or descriptive through the lens of philosophy of science, linking model development to shifts in financial theory and probability paradigms.	Historical Survey	Stock return modeling distinguishes between explanatory and descriptive aims. Since the 1980s, focus shifted toward statistical-inductive models like GARCH to capture patterns such as volatility clustering, emphasizing empirical regularities over causal explanations.
34	**Stock Market**	Atilgan, et al., (2015)	54	Thomson Reuters Web of Science citation database	1983 - 2013	Provides an overview of the growing literature on equity return predictability and volatility.	structured literature review	Few theoretical models address emerging markets, despite their evolution and growing return correlation with developed markets. Structural challenges have prompted partnerships to enhance market infrastructure.
35	**Stock Market**	Broto & Ruiz (2004)	Not disclosed / Specified	Not disclosed / Specified	Not disclosed / Specified	Reviews the literature on estimation methods for stochastic volatility models.	Survey	QML offers ease of use but lower efficiency. Estimation methods like QML, JPR, MCL, and KSC differ in smoothness, with KSC being most volatile. Fixed GMM parameters yield similar results across filters.
36	**Stock Market*** (combine GARCH with …)**	Bhowmik & Wang (2020)	50	Scoups and Web of Science	2008 - 2019	Reviews effective GARCH models for analyzing market returns and volatility, and conducts a content analysis of 50 related studies from 2008 to 2019.	Systematic Literature Review "surveyed by cross-reference snowballing "	Asymmetric GARCH models like EGARCH and TGARCH effectively capture volatility, with EGARCH often performing best. Multivariate GARCH enhances cross-market analysis but is underused. Hybrid approaches improve forecasting, especially during market shifts and crashes.
37	**Stock Market**	Thakkar & Chaudhari (2021)	Not disclosed / Specified	Google Scholar	2011 - 2020	Categorizes fusion into information, feature, and model fusion, reviewing related research in stock price prediction, portfolio management, and risk/return forecasting.	Survey	Fusion techniques enhance non-linear stock prediction, with information and model fusion widely used and feature fusion underexplored. Integration points affect outcomes, and combining fusion types can improve forecasting robustness.
38	**Stock Market**	Subrahmanyam (2010)	Not disclosed / Specified	Not disclosed / Specified	1985 - 2010	Reviews variables used in explaining the cross-section of expected stock returns and highlights unresolved issues in the literature.	Review	Expected returns are shaped by factors like size, value, liquidity risk, momentum, and volatility. Strong past fundamentals and analyst overoptimism often lead to lower future returns.
39	**Stock Market**	Liang, et al., (2013)	Not disclosed / Specified	Not disclosed / Specified	Not disclosed / Specified	Reviews key econophysics methods and examines physical properties of Chinese stock markets, including scaling, correlations, leverage effects, and multifractality.	Survey	The Fractional Market Model aligns well with real market data and aids option pricing, especially in emerging markets. MFDMA outperforms MFDFA in multifractal analysis, while MFDCCA effectively reveals multifractal structures in financial time series.

40	**Stock Market**	Jiang (2021)	115	Google Scholar + arXiv.org	2017 - 2019	Provides a recent review of deep learning models for stock prediction, covering data sources, network structures, evaluation metrics, and reproducibility. It also highlights future research directions to support progress and baseline development.	Survey	KNN and ANN generally outperform SVM in stock prediction, with PCA-ANN offering superior accuracy. Keras and TensorFlow are widely used in deep learning, though such models require GPU support due to high computational demands.
41	**Stock Market**	Sevastianov & Dymova (2009)	Not disclosed / Specified	Not disclosed / Specified	Not disclosed / Specified	Proposes a novel approach combining fuzzy logic and Dempster–Shafer Theory to more naturally model decision-making in stock trading.	systematic review and Empirical Study	DSTmin, based on Fuzzy Dempster–Shafer Theory, proves the most robust and profitable for fuzzy logic-based stock trading, with strong performance across training and testing. Real NYSE data confirms its reliability, even in declining markets.
42	**Stock Market*** power of LSTM**	Kumbure, et al., (2022)	138	Scopus, Web of Science (WoS), IEEE Xplore and Science Direct databases	2000 - 2019	This review extensively examines stock market prediction studies, covering markets, indices, and 2,173 variables (technical, macroeconomic, fundamental). It also analyzes ML techniques used and includes a bibliometric review of influential articles.	Bibliometric analysis and in-depth review	ANNs, SVMs, and fuzzy logic models are widely used in stock prediction, with fuzzy set theory gaining traction. GAs aid optimization, while PCA and wavelets support feature extraction. LSTM has recently outperformed other deep models in prediction robustness.
43	**Stock Market**	Zaznov, et al., (2022)	Not disclosed / Specified	Not disclosed / Specified	Not disclosed / Specified	Critically evaluates current studies on stock price prediction using LOB data, identifying research gaps and suggesting directions for future improvements.	Survey (systematic analysis, comparision, and critical evaluation)	Deep learning models perform well by standard metrics, but profitability often assumes ideal conditions. LSTM remains effective, though CNN-LSTM hybrids improve results. Transformer models adapted to LOB data now outperform LSTM, setting new predictive benchmarks
44	**Stock Market**	Olorunnimbe & Viktor (2022)	35	Google Scholar	2018 - 2020	Surveys deep learning research in stock markets with emphasis on explainability, reproducibility, and practical evaluation. Uniquely adopts backtesting and domain-specific metrics as core assessment criteria.	Systematic survey of practice, backtesting, and applications	DL models like LSTM and DRL are widely used in stock prediction, but online learning remains underutilized. Emphasis is growing on robust backtesting and domain-specific metrics, as DL becomes more viable for real-time financial decision-making.
45	**Stock Market**	Sahoo & Chakraverty (2022)	Not disclosed / Specified	Not disclosed / Specified	Not disclosed / Specified	Provides a comprehensive analysis of ML models for dynamic problems across disciplines, including earthquake prediction, structural dynamics, and stock market forecasting.	A state-of-art review	Highlights the adaptability and effectiveness of ANNs in stock market forecasting, emphasizing their strength in modeling complex, nonlinear patterns and outperforming traditional financial prediction models.
46	**Stock Market**	Alsharef, et al., (2022)	Not disclosed / Specified	Not disclosed / Specified	Not disclosed / Specified	Reviews time-series forecasting challenges and techniques, with a focus on AutoML. Identifies gaps in existing approaches and serves as a reference for researchers and industry practitioners.	Review	Linear models lack predictive accuracy, while DL methods perform better but require expert tuning. AutoML offers a practical solution by automating model selection and optimization for time-series forecasting, streamlining development with minimal manual effort.

47	**Stock Market**	Zhang, et al., (2022)	75	Web of Science (WOS) and Scopus	1995 - 2021	Presents a systematic review of decision fusion models for stock market prediction, analyzing base learners and fusion methods, and categorizing studies by classification or regression tasks.	Systematic Review	Decision fusion models in stock forecasting use both homogeneous and heterogeneous base learners, with classification tasks relying on voting/tree methods and regression on averaging/ANN. The focus has shifted toward decomposition-based ensembles over time.
48	**Stock Market**	Htun, et al., (2023)	32	Web of Science (WOS) and Scopus	2011 - 2022	Reviews feature selection and extraction techniques used in stock market analysis, their integration with ML models, and performance outcomes. Also covers related surveys, input/output data, and factor-based analyses.	Survey	Feature selection is crucial for stock forecasting, with RF, PCA, and AEs commonly used. Most studies rely on single-source inputs, though combining features and applying effective selection methods can greatly improve model accuracy.
49	**Stock Market**	Sabharwal & Aggrawal (2023)	67	Science direct, Springer, IEEE, Wiley and Taylor and Francis.	2012 - 2021	Examines how information spread on social networks influences stock market behavior and forecasting applications.	Survey	Few studies explore how online information influences stock markets in developing countries, with limited attention to its effect on retail investor behavior.
50	**Stock Market**	Janková (2023)	49	Web of Science (WOS) and Scopus	2009 - 2021	Presents a critical review of recent international research on text mining and sentiment analysis for stock market prediction, focusing on stock indices and individual stocks.	Critical Review and Bibliometric Review analysis	Sentiment analysis in finance faces challenges like inconsistent methods, limited data sources, and geographic bias. Improving prediction reliability requires diverse inputs and adaptive models, as inaccurate sentiment scores may misguide investment decisions.
51	**Stock Market**	Cheng, et al., (2022)	N/A	Web of Science (WOS)	2016 - 2021	Summarizes text-based stock forecasting methods—semantic, sentiment, event extraction, and hybrid—evaluating their strengths, limitations, and suitable applications, while outlining future research directions.	Survey Review Analysis	Reviews NLP-based stock forecasting methods—semantic, sentiment, event extraction, and hybrid—highlighting their strengths, limitations, and future directions, such as aspect-based sentiment models and richer data integration for stock correlation analysis
52	**Stock Market*** (news)**	Ashtiani & Raahmei (2023)	61	SCOPUS, WOS, IEEE Xplore, ACM Digital Library, and Science Direct	2015 - 2022	Addresses key questions on data sources, text representation, ML techniques, evaluation metrics, and future research trends in intelligent stock market prediction.	Systematic Literature Review	Literature shows that combining news headlines with numerical data enhances stock prediction, often outperforming full articles. Neural networks, especially BiLSTM and BiGRU, are most effective in extracting market-relevant insights from text.
53	**Stock Market**	Tang, et al., (2022)	146	WOS	2011 - 2021	This review highlights key features of widely used models in financial time series (FTS) prediction, focusing on machine learning applications. It systematically selects recent, highly cited, and strongly connected studies (via bibliographic coupling and co-citation) to guide researchers and investors toward effective model choices and track evolving research trends.	Survey and Review	The literature underscores that robust financial time series (FTS) forecasting models must address noise, nonlinearity, and nonstationarity while capturing both short- and long-term dependencies. Long Short-Term Memory (LSTM) networks are widely favored for their strong sequence modeling and adaptability to large, unstructured data, making them well-suited for modern FTS prediction in the era of big data and AI.

54	**Stock Market**	Kehinde, et al., (2023)	220	SCOPUS	2001 - 2021	This study conducts a scientometric analysis using Scopus data to review 20 years of research on stock market forecasting (SMF). It maps publication trends, collaboration networks, and methodological progress to identify influential studies and research gaps. By combining bibliometric insights with strategic discussion, the study offers a comprehensive overview useful for both academics and practitioners aiming to advance hybrid and data-driven forecasting techniques.	Scientometric review and analysis	The literature highlights strong researcher collaborations and identifies Expert Systems with Applications as the leading journal in stock market forecasting (SMF). Key themes include machine learning, deep learning, and artificial neural networks (ANNs), with fuzzy methods used for clustering. There is a shift toward hybrid and deep learning models. To overcome biases from sentiment and policy in historical data, the study suggests using Inverse Data Envelopment Analysis (DEA) and DEA–ML hybrids. Though limited in scope, the review offers practical insights for researchers, investors, and policymakers.
55	**Energy Market**	Lu, et al., (2021)	171	Google Scholar, Web of Science, Scope, Ei Compendex, Sciencedirect, and IEEE Xplore Digital Library	2010 - 2020	This paper reviews energy price prediction models developed between 2010 and 2020, offering guidance for model selection, typical prediction error ranges, and suggestions for future research directions.	Systematic Review Analysis	Since 2017, energy price forecasting has increasingly focused on hybrid models, with short- to medium-term horizons. Key inputs include environmental and economic factors. Data cleaning methods have a stronger impact on accuracy than optimization techniques
56	**Energy Market**	Ghoddusi, et al., (2019)	130	SCImago Journal Rank (SJR), Science Citation Index Expanded (SCIE), Social Sciences Citation Index (SSCI), and Arts & Humanities Citation Index (A&HCI)	2005 - 2018	The paper reviews recent energy economics literature, identifying commonly used and overlooked methods, underexplored applications, cross-disciplinary insights, and future research opportunities.	Critical Review Analysis	The literature shows crude oil and electricity price forecasting dominate energy finance studies. ANNs remain common; interest in SVMs grows, while DL is underused but promising. Hybrid models like VEC–NARNN, LSSVR-GA, and IDE outperform GARCH, VAR, and ARIMA in accuracy and efficiency. FFNN, ABMs, MARS, and CMARS also show strong performance. The EWH hybrid (ESM–ARIMA–NAR with Kalman Filter) surpasses traditional models.
57	**Energy Market "Crude Oil"**	Chiroma, et al., (2016)	59	Not disclosed / Specified	2001 - 2012	This paper provides a comprehensive critical review of computational intelligence applications in crude oil price forecasting.	In-depth Critical Review	Crude oil price forecasting often uses daily/monthly data. Traditional models aid trend analysis and benchmarking, while wavelet-integrated intelligent models enhance convergence and multi-scale analysis.
58	**Energy Market "Crude Oil"**	Chiroma, et al., (2013)	98	ACM Digital Library, IEEEXplore, Science Direct, Scopus, Springer Link, Web of Science, Google Scholar, Direct Open Access Journals (DOAJ), Microsoft Academic Search,	1980 - 2012	The paper surveys ten years of research on computational intelligence techniques for crude oil price prediction, offering a com	Survey	The literature shows hybrid models like WDNEVaR outperform standalone methods in crude oil price forecasting. ARIMA excels short-term, while ANNs/SVMs suit long-term. WNNs and MLPs also enhance accuracy, supporting a shift toward integrated forecasting models.

				ProQuest and CiteSeerX				
59	**Energy Market "Electricity, Gas, and Oil"**	Todorov & Sánchez Lasheras (2022)	13	The Directory of Open Access Journals, the Multidisciplinary Digital Publishing Institute and the Springer Link.	2017 - 2021	This review examines forecasting methods for stock and commodity prices in electricity, oil, gas, and energy sectors, with a focus on the data variables used to train these models.	Systematic Review Analysis	The review examines recent forecasting methods for stock markets in energy-related sectors, finding only 13 relevant studies out of 1683. It identifies key models and variables, underscoring a research gap and recommending broader, decade-long reviews using major databases.
60	**Energy Market "Electricity price"**	Zema & Sulich (2022)	1294	Scoups and Web of Science	2014 - 2021	The article reviews and compares Electricity Price Forecasting (EPF) models by popularity and accuracy using a bibliometric analysis via VOSviewer. It identifies key gaps and trends, aiming to determine the most cited and most effective EPF methods.	bibliometric study as a variation of Systematic Literature Review (SLR)	The study classifies EPF models, combines bibliometric analysis with SLR, and identifies the most cited and accurate methods. It finds linear models inadequate for capturing pricing asymmetries and recommends focusing on short-term forecasting and standardized keywords for future research.
61	**Currencies "FOREX"**	Pradeepkumar & Ravi (2018)	82	ScienceDirect, IEEE Explore Digital Library, ACM Digital Library, Google Scholar, Springer Link, Taylor & Francis, and Wiley Online Library	1998 - 2017	This paper systematically reviews soft computing models for FOREX rate prediction, identifies research gaps, and highlights the need for hybrid intelligent models to inspire future exploration in underdeveloped areas.	Comprehensive Review Analysis	The literature shows ANN-based hybrids dominate financial forecasting, especially in bankruptcy and risk analysis. Chaos-based, kernel-based, and fuzzy models remain underused in FOREX prediction. While GAs are common, other evolutionary algorithms like DE, ACO, and PSO are underutilized. MLP is prevalent, but advanced and deep learning models are largely unexplored, revealing major gaps and future potential.
62	**Currencies "FOREX"**	Huang, et al., (2004)	Not disclosed / Specified	Not disclosed / Specified	Not disclosed / Specified	This paper attempts to provide a survey of research in forecasting foreign exchange rates with Neural Networks	Survey	The literature shows data normalization offers inconsistent benefits in neural network forecasting. Smaller training sets often improve results and reduce costs. While feedforward ANNs like MLPs model complex patterns quickly, RNNs are better for exchange rate data. Forecast accuracy hinges on input choice, data prep, and architecture, but ANN design remains uncertain due to its nonparametric nature.
63	**Currencies "FOREX"**	Ayitey Junior, et al., (2023)	60	Googlee Schoar, IEE Xplore digital Library, ScienceDirect, Microsoft Academic, and ACM digital Library	2010 - 2021	The objective is to give an overview of machine learning models and their application in the FX market.	Systematic Literature Review and Meta Analysis	A 2010–2021 review of 60 studies found deep learning models (e.g., ANN, LSTM, MLP, RBF) outperform traditional ML models (e.g., KNN, Naïve Bayes) in forex forecasting. It also identifies key validation practices and highlights unresolved challenges for future research.

64	**Currencies "FOREX"**	Fang, et al., (2023)	4999	Web of Science (WOS) Core collection database	1992-2021	This paper uses bibliometric analysis via CiteSpace to map exchange rate research across economics, finance, and related fields, revealing publication, author, and journal networks while offering insights for future research directions.	Bibliometric Review Analysis	The reviewed literature reveals a shift in exchange rate forecasting from traditional econometric models (ARIMA, VAR, GARCH) to AI-based methods (ANN, SVM, fuzzy systems), driven by major global events and technological advances. Leading contributions come from the U.S., U.K., and China. The integration of big data, NLP, and hybrid models marks a key future direction for enhancing predictive accuracy.
65	**Currencies "Cryptocurrencies"**	Bariviera & Merediz-Solà (2021)	**438** "bibliometric analysis" and **106** "in-depth literature review"	Web of Science	Not disclosed / Specified	This survey uses a hybrid bibliometric and literature review to provide an updated overview of cryptocurrency research in economics.	Survey based on hybrid analysis: bibliometric analysis and close literature review	The literature views cryptocurrencies as speculative tools for high-frequency analysis, but research remains Bitcoin-centric. Broader token studies, blockchain environmental impact, mining protocol effects, and regulatory theory are underexplored. Rising central bank interest calls for interdisciplinary research.
66	**Currencies "Cryptocurrencies"**	Wątorek, et al., (2020)	Not disclosed / Specified	Not disclosed / Specified	Not disclosed / Specified	This review examines whether the complexity of cryptocurrency exchange rates now resembles that of traditional markets (stocks, bonds, commodities, currencies) following rapid development.	Survey	The literature identifies cryptocurrencies—mainly Bitcoin—as key tools for studying high-frequency dynamics, though broader token analysis remains limited. Economic studies overlook blockchain's environmental impact, mining protocol effects, and regulatory theory. With central banks exploring digital currencies, interdisciplinary research is increasingly essential.
67	**Currencies "Cryptocurrencies"**	Fang, et al., (2022)	146	Google Scholar and arXiv	2013 - June 2021	The literature is categorized into six areas: trading systems, systematic strategies, emerging methods (e.g., ML), portfolio and comovement studies, market condition analyses (e.g., bubbles/crashes), and other topics. The survey compiles key studies and uses descriptive indicators to assess research maturity in cryptocurrency trading.	Comprehensive Survey	The reviewed literature surveys 146 cryptocurrency trading studies, providing a taxonomy, key datasets, and emerging trends. It suggests future research on advanced NLP, trading horizons, asset correlations, and models like game theory and agent-based systems. It also stresses balancing research transparency with the risk of reduced trading advantage.
68	Financial Markets	Current Study	67 (SLR) + 4982 (Bibliometric)	WOS	1 Jan. 1990 – 31 Dec. 2024	To systematically review prior review-based studies on modeling in financial markets (Stage 1), followed by large-scale bibliometric analysis (Stage 2) and content synthesis (Stage 3) to map themes, assess methodologies, and guide future research.	Comprehensive: 3-stage: SLR (PRISMA) → Bibliometric Analysis (Biblioshiny	This study uniquely aggregates and analyzes review literature only in Stage 1 (67 SLRs), followed by extensive bibliometric (4982 records) and content analysis. It highlights fragmentation, lack of integrated themes, and methodological limitations in past reviews. The multi-stage structure offers a unified, future-oriented research framework.

							) → Content Analysis (Thematic)	

The results of SLR revealed significant fragmentation and methodological gaps in the existing review for FMM. The first implication of SLR findings is that most existing review studies are conceptually descriptive or methodologically narrow. Over 70% of the papers adopted narrative styles lacking transparency, reproducibility and theoretical synthesis (Weyori et al., 2021; Kumar et al., 2021). These reviews focused on individual model families or specific markets, but didn't comprehensively explore cross-model comparisons, domain interactions, or thematic evolutions over time. Even papers employing bibliometric techniques as Baker et al. (2020) failed to triangulate structural findings with qualitative interpretation. So, SLR revealed an urgent need for a study that integrates **systematic, bibliometric and content analyses** capable of answering questions about how forecasting models are developed, applied, and compared across financial domains.

SLR not only uncovered the gaps but also directly shaped the bibliometric design. Most prior reviews relied on multi-database searches but rarely justified their choices. In contrast, this study purposefully chose **WOS** as the sole database for bibliometric analysis due to its reliability, robust citation tracking, compatibility with advanced tools and relevance in high-impact finance which is aligns with Aria and Cuccurullo (2017). Based on SLR's finding that reviews were limited in scale, the bibliometric analysis expanded the corpus to **4,982 research papers which is** allows for statistically valid co-citation, co-authorship, keyword co-occurrence, and thematic mapping, thus offering a macro-level view of the intellectual and conceptual evolution of FMM. SLR confirmed that very few review studies utilized advanced bibliometric software. **Biblioshiny for Bibliometrix** (RStudio) enables nuanced exploration of productivity, influence, collaboration networks, and thematic trends, all while addressing the research questions identified in SLR (Kumar et al., 2021; Baker et al., 2020).

As bibliometric analysis addressed the structural and macro patterns, SLR revealed the need for micro-level interpretive strategy. This droves the decision to conduct a **content analysis** of selected articles derived from stage 2's citation and thematic clusters. SLR found that many reviews did not interpret models within their applied contexts or market conditions. Hence, we adopted **content analysis** to examine the methodologies, data types, market domains and application frameworks of the most-cited papers in each cluster. SLR highlighted that no previous review attempted a three-dimensional analysis of **model-type × market-type × time-horizon** effectiveness. Thus, content analysis was tasked with identifying patterns within this multi-layered matrix, particularly focusing on neural models, hybrid AI approaches and statistical benchmarks (Ballings et al., 2015; Zhang et al., 2022). SLR provided clarity on **what was missing:** lack of consensus on model evaluation criteria: some papers used accuracy metrics; others emphasized profitability or volatility forecasts which is motivated a comparative framework in stage 2 and 3. Neglect of alternative markets: while stocks and cryptocurrencies dominated recent reviews, ESG-related markets, and macroeconomic indexes were underexplored. Stage 2's keyword design and stage 3's content selection addressed this imbalance. Disconnection between structural patterns and practical applications: SLR's identification of this disjoint led to a **triangulated design**, where bibliometric informs content depth, and both are anchored in a systematic foundation.

Table S2: General Information and Performance analysis for FMM

	Description	Results		Description	Results
Main information	Timespan	1990:2024	Authors	Authors	9,468
	Sources (Journals, Books, etc)	758		Authors of single-authored docs	637
	Documents	4,982[1]	Authors' collaboration	Single-authored docs	752
	Annual Growth Rate %	9.5%		Co-Authors per Doc	2.79
	Document Average Age	8.4		International co-authorships %	26.82
	Average citations per doc	26.72			
	References	101,315			

[1] Two articles out of the 4,984 were excluded as they were classified as early access publications from 2025, whereas the review strictly focuses on literature published up to the end of 2024.

Document contents	Keywords Plus (ID)	3,495
	Author's Keywords (DE)	10,205

Table S3: The highest contributors in FMM (Top 20 contributors)

Authors	Articles		Articles Fractionalized	Authors	Articles		Articles Fractionalized
	No	%			No	%	
GUPTA R	63	11.70%	20.23	CHEN Y	19	3.50%	6.16
MA F	60	11.20%	16.20	SALISU AA	19	3.50%	6.67
ZHANG YJ	47	8.85	13.83	WANG JQ	19	3.50%	5.15
WANG J	38	7%	14.33	NARAYAN PK	18	3.40%	8.17
WANG YD	31	5.80%	9.50	ZHANG X	18	3.40%	4.52
PIERDZIOCH C	25	4.60%	8.33	LIU J	17	3.10%	4.79
WANG L	25	4.60%	7.20	MCMILLAN DG	17	3.10%	13.25
LI Y	24	4.40%	7.03	BOURI E	16	3%	4.45
LIANG C	23	4.30%	6.07	WANG Y	16	3%	3.97
WANG SY	23	4.30%	6.77	WEI Y	16	3%	4.18

Table S4: Most Local Cited Authors and author Production Over Time

	Author	Local Citations			Author	Local Citations	
		NO	%			NO	%
Top Cited Authors	MA F	549	9.3%	Other notable contributors	ENKE D	146	2.50%
	CAMPBELL JY	433	7.30%		JIANG FW	146	2.50%
	ZHANG YJ	338	5.80%		ENKE D	146	2.50%
	THOMPSON SB	313	5.30%		RAPACH DE	144	2.40%
	GUPTA R	255	4.30%		BEKAERT G	143	2.40%
Mid-Tier Cited Authors	WEI Y	212	3.60%		LUX T	136	2.30%
	NARAYAN PK	205	3.50%		PATEL J	136	2.30%
	ZHOU GF	187	3.20%		SHAH S	136	2.30%
	WANG YD	185	3.10%		THAKKAR P	136	2.30%
	WANG J	183	3.10%		TU J	133	2.20%
	WANG SY	173	3.00%		LIU J	126	2.10%
Emerging and Specialized Contributors	KIM HY	169	2.80%		KIM KJ	125	2.10%
	HUANG DS	162	2.70%		ANG A	124	2.10%
	LIANG C	160	2.70%		BOYACIOGLU MA	124	2.10%
	ATSALAKIS GS	156	2.60%		LI Y	124	2.10%
	KOTECHA K	156	2.60%		Total	5861	100%

Table S5: Authors' Local Impact: Top 20 most publishing and prominent authors

	Author	H-index	G-index	M-index	TC	NP	PY-start
Leading Authors	MA F	26	41	2.6	1,788	60	2015
	GUPTA R	23	46	1.769	2,146	63	2012
High contributors	WANG J	19	32	0.95	1,081	38	2005
	ZHANG YJ	19	33	1.267	1,131	47	2010
	NARAYAN PK	16	18	0.842	1,325	18	2006

Strong local impact	LIANG C	14	23	2.8	605	23	2020
	WANG YD	14	27	1.077	768	31	2012
	BOURI E	13	16	1.625	1,240	16	2022
	WEI Y	13	16	0.765	714	16	2008
Emerging contributors	LI Y	11	22	0.917	501	24	2013
	LIU J	11	17	0.524	558	17	2004
	WANG SY	11	23	0.524	1,116	23	2004
	WU CF	11	11	0.786	559	11	2011
	BALCILAR M	10	14	0.833	824	14	2013
	CHEN W	10	15	1.25	512	15	2017
	CHENG CH	10	14	0.556	566	14	2007
	LEE J	10	15	0.556	676	15	2007
	PIERDZIOCH C	10	14	0.455	230	25	2003
	SALISU AA	10	19	1.429	592	19	2018
	SORNETTE D	10	15	0.385	681	15	1999
	WANG JQ	10	19	2	412	19	2020

Table S6: Top Journals by Article Count in FMM

Sources	Articles	Sources	Articles
PHYSICA A-STATISTICAL MECHANICS AND ITS APPLICATIONS	190	APPLIED SOFT COMPUTING	57
EXPERT SYSTEMS WITH APPLICATIONS	179	PLOS ONE	56
JOURNAL OF FORECASTING	145	EMERGING MARKETS FINANCE AND TRADE	55
ENERGY ECONOMICS	120	JOURNAL OF INTERNATIONAL MONEY AND FINANCE	54
FINANCE RESEARCH LETTERS	116	APPLIED ECONOMICS LETTERS	53
APPLIED ECONOMICS	81	JOURNAL OF FUTURES MARKETS	53
ECONOMIC MODELLING	81	INTERNATIONAL JOURNAL OF FINANCE & ECONOMICS	49
IEEE ACCESS	81	ENERGIES	47
INTERNATIONAL REVIEW OF FINANCIAL ANALYSIS	72	ECONOMICS LETTERS	46
INTERNATIONAL REVIEW OF ECONOMICS & FINANCE	68	ANNALS OF OPERATIONS RESEARCH	45
INTERNATIONAL JOURNAL OF FORECASTING	65	SOFT COMPUTING	44
COMPUTATIONAL ECONOMICS	64	PACIFIC-BASIN FINANCE JOURNAL	40
NORTH AMERICAN JOURNAL OF ECONOMICS & FINANCE	63	JOURNAL OF EMPIRICAL FINANCE	38
QUANTITATIVE FINANCE	63	RESOURCES POLICY	38
MATHEMATICS	61	Other Sources (728 Journals)	2,799
JOURNAL OF BANKING & FINANCE	59		
Total	4,982		

Table S7: Most Cited (Local) Sources

Sources	Articles	%	Sources	Articles	%
J FINANC	8,546	12%	*APPL SOFT COMPUT*	1,407	1.90%
J FINANC ECON	6,102	8.60%	*J FORECASTING*	1,405	1.90%
EXPERT SYST APPL	4,880	6.80%	*J EMPIR FINANC*	1,393	1.90%
REV FINANC STUD	4,743	6.60%	J BUS	1,377	1.90%
ENERG ECON	4,293	6.2%	*J FINANC QUANT ANAL*	1,337	1.80%
J ECONOMETRICS	4,124	5.80%	*ECON MODEL*	1,242	1.70%
ECONOMETRICA	3,816	5.30%	*NEUROCOMPUTING*	1,185	1.60%
PHYSICA A	3,229	4.60%	*INT REV FINANC ANAL*	1,181	1.60%
J BANK FINANC	2,448	3.40%	*ECON LETT*	1,122	1.50%
INT J FORECASTING	2,274	3.20%	*J FUTURES MARKETS*	1,117	1.50%
J BUS ECON STAT	2,062	3.1%	*J ECON DYN CONTROL*	1,067	1.50%
FINANC RES LETT	1,812	2.60%	*REV ECON STAT*	1,046	1.50%
AM ECON REV	1,701	2.40%	*J AM STAT ASSOC*	1,038	1.50%
J POLIT ECON	1,462	2.2%	*MANAGE SCI*	1,001	1.40%
J INT MONEY FINANC	1,448	2.1%	*ENERGY*	981	1.40%

Table S8: Top 30 Journals by H-Index Impact

Source		H_index	G_index	M_index	TC (N)	TC (%)	NP	PY_start
EXPERT SYSTEMS WITH APPLICATIONS	Top Contributors	63	113	2.423	13297	18.7%	179	1999
ENERGY ECONOMICS		45	72	1.286	5590	7.8%	120	1990
PHYSICA A-STATISTICAL MECHANICS AND ITS APPLICATIONS		35	51	1.346	4086	5.7%	190	1999
APPLIED SOFT COMPUTING	Mid-Tier Contributors	30	52	1.667	2758	3.8%	57	2007
JOURNAL OF FORECASTING		29	43	0.879	2443	3.4%	145	1992
INTERNATIONAL JOURNAL OF FORECASTING		28	54	0.824	2968	4.1%	65	1991
ECONOMIC MODELLING		25	41	0.833	1893	2.6%	81	1995
JOURNAL OF BANKING & FINANCE		25	46	0.714	2207	3.1%	59	1990
JOURNAL OF INTERNATIONAL MONEY AND FINANCE	Specialized and Emerging Sources	24	49	0.686	2435	3.4%	54	1990
IEEE ACCESS		23	41	3.286	1839	2.5%	81	2018
FINANCE RESEARCH LETTERS		22	40	1.467	1840	2.5%	116	2010
INTERNATIONAL REVIEW OF FINANCIAL ANALYSIS		20	36	1.667	1467	2.0%	72	2013
NEUROCOMPUTING		20	27	0.69	1340	1.8%	27	1996
JOURNAL OF ECONOMETRICS		19	27	0.543	2900	4.0%	27	1990
JOURNAL OF ECONOMIC DYNAMICS & CONTROL		18	26	0.529	702	0.009%	30	1991
JOURNAL OF EMPIRICAL FINANCE		18	34	1.059	1204	1.6%	38	2008
JOURNAL OF FUTURES MARKETS		18	35	0.514	1289	1.8%	53	1990
NORTH AMERICAN JOURNAL OF ECONOMICS & FINANCE		18	32	1.5	1130	1.5%	63	2013
QUANTITATIVE FINANCE		18	27	0.818	907	1.2%	63	2003
REVIEW OF FINANCIAL STUDIES		18	22	0.563	5129	7.2%	22	1993

APPLIED ECONOMICS		17	26	0.486	855	1.25	81	1990
ENERGY		17	28	1.063	904	1.25	28	2009
JOURNAL OF FINANCE		17	17	0.486	3353	4.7%	17	1990
JOURNAL OF FINANCIAL ECONOMICS		17	18	0.515	4172	5.8%	18	1992
ENERGIES		16	23	1.333	616	0.008%	47	2013
NEURAL COMPUTING & APPLICATIONS		16	29	0.552	909	1.2%	29	1996
PACIFIC-BASIN FINANCE JOURNAL		16	29	1	856	1.25	40	2009
PLOS ONE		16	32	1.231	1098	1.55	56	2012
RESOURCES POLICY		16	29	1.6	898	1.2%	38	2015

Table S9: Most Cited Documents by Total and Local Citations

Paper		Local Citations	Total Citations	TC per Year	Normalized TC
BOLLEN J, et al., 2011, J COMPUT SCI-NETH	Most Cited Documents based on TC	78	1401	100.07	29.37
CAMPBELL JY, & THOMPSON, SB., 2008, REV FINANC STUD		313	1388	81.65	36.81
LUX T, & MARCHESI, M., 1999, NATURE		104	997	38.35	13.93
FISCHER T, & KRAUSS, C., 2018, EUR J OPER RES		111	976	139.43	27.81
CAMPBELL JY, & HENTSCHEL, L., 1992, J FINANC ECON		32	972	29.45	11.43
CHEN J, ET AL., 2001, J FINANC ECON		14	941	39.21	15.69
KONNO H, & YAMAZAKI, H., 1991, MANAGE SCI		6	922	27.12	11.95
ANG A, & BEKEART, G., 2007, REV FINANC STUD		124	658	36.56	16.66
CLAUS J, & THOMAS, J., 2002, J FINANC		8	648	27.00	10.80
HUANG W, ET AL., 2005, COMPUT OPER RES		98	605	30.25	11.07
MCLEAN RD, & PONTIFF, J., 2016, J FINANC		20	599	66.56	18.62
BOLLERSLEV T, & MIKKELSEN, H.O., 1996, J ECONOMETRICS		43	581	20.03	8.29
KATSIAMPA P, 2017, ECON LETT		68	577	72.13	18.22
MORRIS S., & SHIN, H. S., 1998, AM ECON REV		9	571	21.15	10.83
HUANG DS, ET AL., 2015, REV FINANC STUD		94	571	57.10	14.69
PAI PF, & LIN, CS., 2005, OMEGA-INT J MANAGE S		69	559	27.95	10.23
PAGAN AR, & SCHWERT, GW., 1990, J ECONOMETRICS		43	545	15.57	9.14
PATEL J, ET AL., 2015, EXPERT SYST APPL-a		82	517	51.70	13.30
VERONESI P, 1999, REV FINANC STUD		17	475	18.27	6.63
ATSALAKIS GS., & VALAVANIS, KP., 2009, EXPERT SYST APPL		94	474	29.63	11.40

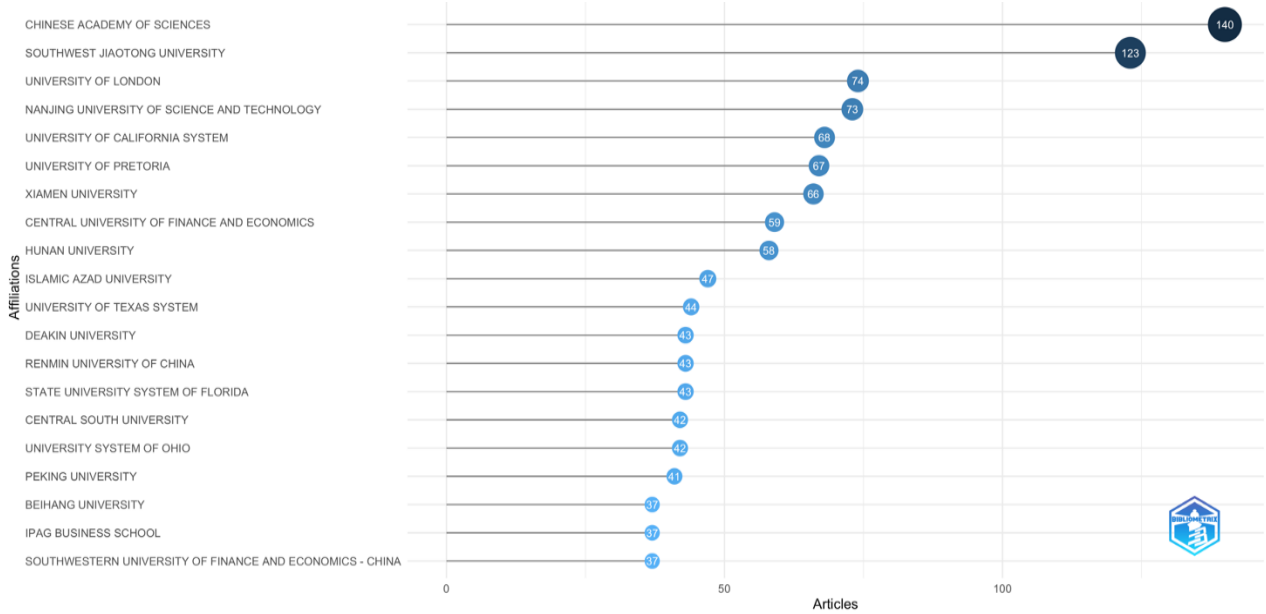

Figure S1: Most Relevant Affiliations

Table S10: Leading Countries by Publications and Citations in FMM Research

Country	Articles		Frequency	Classes based on Frequency	SCP	MCP	MCP %	TC	Classes based on TC
	No	%							
CHINA	1,472	29.5	3,306	Top Contributor	1,213	259	17.6	27,554	Rapidly Rising Competitor
USA	635	12.7	1299	Major Contributors	479	156	24.6	36,694	Dominant Global Leader
UK	276	5.5	656		169	107	38.8	6,991	Leading European Contributors
GERMANY	202	4.1	391		151	51	25.2	6,055	
KOREA	201	4	378		156	45	22.4	5,596	Significant Regional Players
AUSTRALIA	188	3.8	347	Moderate Contributors	108	80	42.6	4,926	
INDIA	187	3.8	407		164	23	12.3	4,518	Emerging Nations
ITALY	134	2.7	302		91	43	32.1	2,260	Mid-Tier European Nations
JAPAN	108	2.2	216		93	15	13.9	2,535	Others in the top 20
FRANCE	105	2.1	325		40	65	61.9	2,230	Mid-Tier European Nations
IRAN	100	2	212		74	26	26	2,733	Emerging Nations
CANADA	95	1.9	232		55	40	42.1	2,810	
BRAZIL	73	1.5	163	Emerging and Sparse Contributors	58	15	20.5	1,612	Other Countries in the Top 20
SPAIN	71	1.4	183		52	19	26.8	1,893	
TURKEY	64	1.3	157		50	14	21.9	2,568	
POLAND	63	1.3	111		53	10	15.9	1,389	

SWITZERLAND	48	1	113		30	18	37.5	1,990	Mid-Tier European Nations
SAUDI ARABIA	46	0.9	142		19	27	58.7	-------	Other Countries
GREECE	45	0.9	-------		31	14	31.1	1,373	
MALAYSIA	42	0.8	106		22	20	47.6	-------	